AlphaEval: Evaluating Agents in Production

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Abstract

The rapid deployment of AI agents in commercial settings has outpaced the development of evaluation methodologies that reflect production realities. Existing benchmarks measure agent capabilities through retrospectively curated tasks with well-specified requirements and deterministic metrics—conditions that diverge fundamentally from production environments where requirements contain implicit constraints, inputs are heterogeneous multi-modal documents with information fragmented across sources, tasks demand undeclared domain expertise, and success is judged by domain experts. We present *AlphaEval*, a production-grounded benchmark of **94** tasks sourced from **seven** companies deploying AI agents in their core business, spanning six O*NET occupational domains. Unlike model-centric benchmarks, *AlphaEval* evaluates *complete agent products*—Claude Code, Codex, etc.—as commercial systems, capturing performance variations invisible to model-level evaluation. Our evaluation framework covers multiple paradigms (LLM-as-Judge, reference-driven metrics, formal verification, rubric-based assessment, and automated UI testing), with individual domains composing multiple paradigms. Beyond the benchmark itself, we contribute a *requirement-to-benchmark construction framework*—a systematic methodology that transforms authentic production requirements into executable evaluation tasks in minimal time. Given any real-world production requirement, this framework enables rapid construction of fully automated, reproducible evaluations—making production-level agent assessment as accessible as research benchmarking.

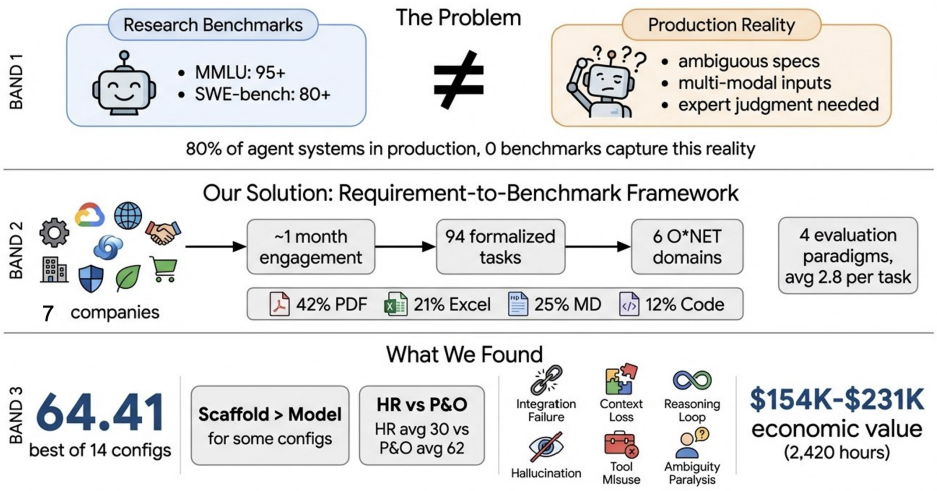

Figure 1: Overview of *AlphaEval*. **Band 1**: The gap between research benchmarks and production reality. **Band 2**: Our requirement-to-benchmark framework transforms production requirements into 94 formalized tasks across 6 O*NET domains. **Band 3**: Key findings—best agent scores 64.41/100, scaffold matters as much as model, and six production-specific failure modes.

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https://www.onetonline.org/find/descriptor/browse/2.A

1 Introduction

The hardest part of deploying an AI agent is not building the agent—it is knowing whether the agent works. As organizations move from prototypes to production, constructing reliable evaluation infrastructure has emerged as a dominant engineering challenge (Pan et al., 2025). Yet the research community has devoted remarkably little attention to *evaluating agents as they are actually deployed in production*. Existing benchmarks like SWE-bench (Jimenez et al., 2024), WebArena (Zhou et al., 2024), and OSWorld (Xie et al., 2024) have been instrumental in driving agent capabilities forward. However, they share a common design philosophy: tasks are curated retrospectively from existing artifacts with well-specified requirements and deterministic metrics—diverging fundamentally from production evaluation. A recent systematic analysis of 43 benchmarks covering 72,342 tasks confirms this mismatch quantitatively, revealing that agent development remains heavily programming-centric and misaligned with the occupational categories where human labor and economic value concentrate (Wang et al., 2026).

Production evaluation operates under different constraints: requirements contain implicit constraints and arrive as loosely worded business descriptions; inputs are heterogeneous multi-modal documents with information fragmented across sources; tasks demand undeclared domain expertise; and success is judged by domain experts' subjective assessments. Despite over 80% of surveyed practitioners reporting their agent systems in production or pilot phases (Pan et al., 2025), no benchmark captures this reality. The disconnect is structural along three axes: (1) *Task Under-specification*: benchmarks have explicit goals; production tasks emerge from evolving business needs with implicit constraints invisible to outsiders. (2) *Judgment Subjectivity*: benchmarks use automatable metrics; production requires domain expertise. (3) *Continuous Evolution*: benchmarks are built once; production evaluations must continuously evolve.

We address this by partnering with companies deploying AI agents in their core business and AI-focused organizations to capture authentic evaluation scenarios. The result is *AlphaEval*, a benchmark of **94** production-grounded tasks together with a reusable *requirement-to-benchmark construction framework* that establishes the shortest path from authentic production requirements to fully automated evaluations. We evaluate six frontier models—Claude Opus 4.6, GPT-5.2, Gemini 3 Pro Preview, Kimi K2.5, GLM-5, and MiniMax M2.5—deployed through four commercial agent products: Claude Code, Codex, GitHub Copilot, and Cursor, yielding 14 model–scaffold configurations. Our key findings: (1) the best configuration (Claude Code + Opus 4.6) achieves only 64.41/100, revealing a substantial research-production gap; (2) scaffold choice matters as much as model choice—the same Opus 4.6 model scores 64.41 via Claude Code but only 53.45 via Codex, an 11-point spread; and (3) domain performance varies dramatically, from 62.0 (Technology Research) to 30.0 (Human Resources), indicating that no single aggregate score captures production readiness (Figure 1).

Our contributions are fourfold:

- **A requirement-to-benchmark construction framework.** A systematic methodology that transforms authentic production requirements from companies into fully automated, reproducible evaluation tasks. This establishes the shortest path from real-world production needs to executable benchmarks—any production requirement can be rapidly operationalized into a rigorous, automated evaluation without compromising fidelity to the original business context.
- **A production-grounded agent benchmark.** 94 tasks from seven companies classified into six O*NET occupational domains, preserving production complexity—ambiguous specifications, multi-modal inputs, and domain-specific evaluation criteria tied to customer satisfaction.
- **A unified evaluation framework.** Multiple evaluation paradigms through a standardized API with Docker-based sandboxed execution. Individual domains *compose* multiple paradigms, reflecting the multi-dimensional nature of production quality.
- **An empirical analysis of the research-production gap.** Frontier agents exhibit significant degradation on production tasks, with the best configuration achieving only 64.41 on average. We conduct a systematic error analysis of failure cases.

2 Preliminaries

We systematically analyze 90+ existing benchmark (agent-relevant) to abstract taxonomy of these evaluation methodologies. As illustrated in Figure 2, multiple evaluation paradigms are summarized: **Reference Answer Verification** (∼11%, 10 tasks) via string and semantic matching; **Formal Logic Verification** (∼21%, 20+ tasks) for mathematical proofs and code analysis; **Rubric-based Evaluation** (∼56%, 53 tasks) with manual, semi-automated, and human-AI co-designed rubrics; and **Execution-based Verification** (∼12%, 11 tasks) via code unit testing

Open-source at `https://github.com/GAIR-NLP/AlphaEval`.

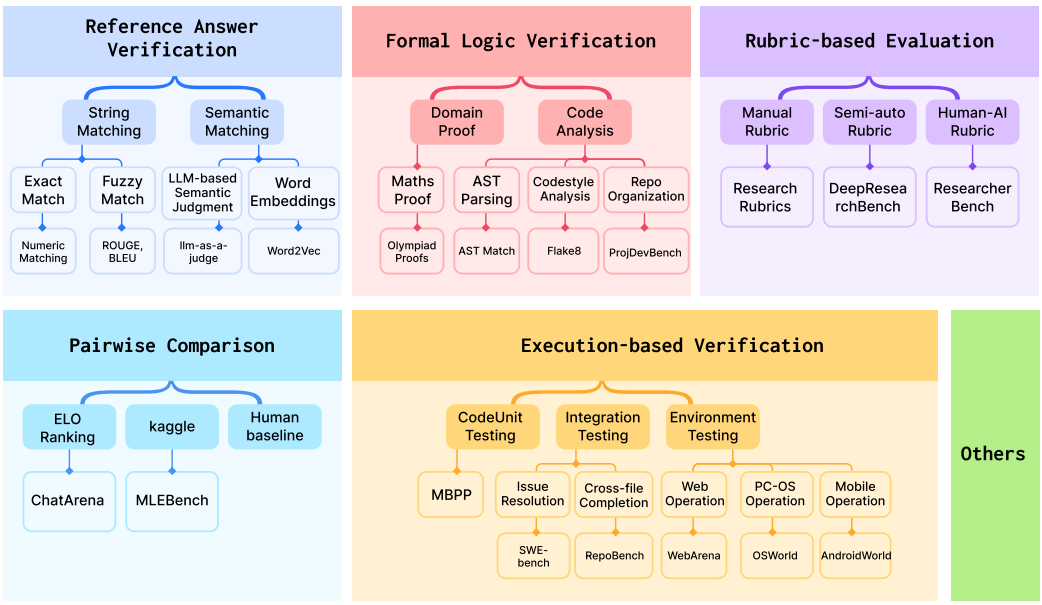

Figure 2: Taxonomy of evaluation methodologies for AI agents. *AlphaEval* covers multiple evaluation paradigms (excluding Pairwise Comparison and Others).

and environment state testing (detailed coverage in Table 6). We additionally employ LLM-as-Judge (Claude Opus 4.6; ∼56%, 53 tasks) as a cross-cutting semantic evaluation method within multiple paradigms. We exclude only Pairwise Comparison (ELO ranking, competition-style) and Others. Critically, individual domains *compose* multiple paradigms—e.g., clinical research combines LLM-as-Judge with numerical verification. Every task employs ≥2 leaf-node evaluation types (avg. 2.8; Table 8), reflecting the multi-dimensional nature of production quality. All rubric scripts output standardized scores (0.0–1.0) enabling cross-domain comparison.

Revisit Existing Agent Benchmarks Task-based agent benchmarks (Jimenez et al., 2024; Zhou et al., 2024; Xie et al., 2024; Li et al., 2026a) enable rigorous capability measurement but curate tasks retrospectively with automatable metrics. Code benchmark surveys (Cao et al., 2025) and standardized evaluation benchmarks (Luo et al., 2025) pursue evaluation rigor but remain anchored in research contexts. Domain-specific evaluation for finance (Kuang et al., 2025), e-commerce (Min et al., 2025), and healthcare (Wang et al., 2025a) remain researcher-curated with predetermined scopes. The rise of LLM-as-Judge enables scalable evaluation but introduces reliability challenges (Panickssery et al., 2024; Zheng et al., 2023) exacerbated in production settings. Notably, production evaluation spans diverse paradigms—LLM-as-Judge, reference-driven metrics, formal verification, rubric-based assessment, and automated UI testing—with most domains combining multiple paradigms, a diversity that existing benchmarks do not capture. Concurrently, Wang et al. (2026) systematically map 43 agent benchmarks to all 1,016 U.S. occupations via O*NET, finding substantial coverage gaps and proposing coverage, realism, and granular evaluation as design principles—criteria that *AlphaEval* operationalizes through production-sourced tasks across six O*NET domains and multi-paradigm evaluation. *AlphaEval* addresses this gap by systematizing production evaluation from companies deploying AI agents, preserving production complexity and contributing both a benchmark and a requirement-to-benchmark construction framework. A comprehensive comparison with 90+ benchmarks across seven dimensions is provided in Table 1.

Table 1: Comprehensive comparison of *AlphaEval* with existing benchmarks across seven dimensions that characterize production-level evaluation: **Prod.** = tasks sourced from real commercial deployments with paying customers; **M.M.** = requires processing multi-modal heterogeneous inputs (PDFs, spreadsheets, images); **Under.** = tasks deliberately preserve production-level requirement ambiguity; **Div.E.** = employs three or more distinct evaluation paradigms; **Exp.V.** = evaluation criteria co-developed and validated with domain experts; **Dyn.** = benchmark designed for continuous evolution rather than static one-time release; **Cross.** = spans three or more distinct professional domains.

Benchmark	Year	Domain	Eval Type	Prod.	M.M.	Under.	Div.E.	Exp.V.	Dyn.	Cross.
Software Engineering & Coding										
SWE-bench (Jimenez et al., 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✗	✗
SWE-bench Multimodal (Yang et al., 2024)	2024	Code	Rule	✗	✓	✗	✗	✗	✗	✗
Multi-SWE-bench (Zan et al., 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
SWE-Lancer (Miserendino et al., 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✗	✗
SWT-Bench (Mundler et al., 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✗	✗
Terminal-bench (Laude Institute, 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
FeatBench (Chen et al., 2025b)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
DevBench (Li et al., 2024)	2024	Code	R+M+H	✗	✗	✗	✓	✗	✗	✗
LongCLI-Bench (Feng et al., 2026)	2026	Code	Rule	✗	✗	✗	✗	✗	✗	✗
ProjDevBench (Lu et al., 2026)	2026	Code	Rule+Model	✗	✗	✗	✗	✗	✗	✗
Data Science & ML Engineering										
DSBench (Jing et al., 2024)	2024	Code	Model	✗	✓	✗	✗	✗	✗	✗
MLE-bench (Chan et al., 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✗	✗
KernelBench (Ouyang et al., 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✗	✗
DAComp (Lei et al., 2025)	2025	Code+Research	Rule+Model	✗	✗	✗	✗	✗	✗	✗
Code Competition & Security										
LiveCodeBench (Jain et al., 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✓	✗
CodeElo (Quan et al., 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
Aider Polyglot (Gauthier, 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✗	✗
CyBench (Zhang et al., 2024)	2024	Code	Rule	✗	✗	✗	✗	✗	✗	✗
BountyBench (Zhang et al., 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✗	✗
VimGolf-Gym (Cybergod AGI Research, 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✗	✗
DPAI Arena (JetBrains, 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
Spring AI Bench (Spring Community, 2025)	2025	Code	Rule	✗	✗	✗	✗	✗	✓	✗
AGENTS.md Eval (Gloaguen et al., 2026)	2026	Code	Rule	✗	✗	✗	✗	✗	✗	✗
Tool Use & Web Interaction										
WebArena (Zhou et al., 2024)	2024	Web	Rule+Model	✗	✗	✗	✗	✗	✗	✓
AgentBench (Liu et al., 2023)	2023	Tool	Rule	✗	✗	✗	✗	✗	✗	✓
AgentBoard (Ma et al., 2024)	2024	Tool	Rule	✗	✗	✗	✗	✗	✗	✓
τ-bench (Yao et al., 2024)	2024	Tool	Rule	✗	✗	✗	✗	✗	✗	✗
τ^2-Bench (Barres et al., 2025)	2025	Tool	Rule	✗	✗	✗	✗	✗	✗	✗
TheAgentCompany (Xu et al., 2024)	2024	Tool	Rule+Model	✗	✓	✗	✗	✗	✗	✓
Tool Decathlon (Li et al., 2025)	2025	Tool	Rule	✗	✗	✗	✗	✗	✗	✓
ACEBench (Chen et al., 2025a)	2025	Tool	Rule+Model	✗	✗	✗	✓	✗	✗	✓
MCP-Universe (Luo et al., 2025)	2025	Tool	Rule	✗	✗	✗	✗	✗	✓	✓
BFCL (Patil et al., 2025)	2025	Tool	Rule	✗	✗	✗	✗	✗	✓	✓
Context-Bench (Letta, 2025a)	2025	Tool	Rule	✗	✗	✗	✗	✗	✓	✗
Letta Evals (Letta, 2025b)	2025	Tool	Rule+Model	✗	✗	✗	✗	✗	✓	✗
EcomBench (Min et al., 2025)	2025	Tool	Model	✗	✗	✗	✗	✓	✓	✗
DeliveryBench (Mao et al., 2025)	2025	Tool	Rule	✗	✓	✗	✗	✗	✗	✗
WorFBench (Qiao et al., 2024)	2024	Tool	Rule	✗	✗	✗	✗	✗	✗	✓
BrowseComp (Wei et al., 2025)	2025	Search	Model	✗	✗	✗	✗	✗	✗	✓
AgencyBench (Li et al., 2026a)	2026	Code+Tool	Rule+Model	✗	✓	✗	✗	✓	✗	✓
Operating System & GUI										
GAIA (Mialon et al., 2023)	2023	OS	Rule	✗	✓	✗	✗	✗	✗	✓
OSWorld (Xie et al., 2024)	2024	OS	Rule	✗	✓	✗	✗	✗	✗	✗
AppWorld (Trivedi et al., 2024)	2024	OS	Rule	✗	✗	✗	✗	✗	✗	✗
WebSuite (Li and Waldo, 2024)	2024	OS	Rule	✗	✗	✗	✗	✗	✗	✗
HammerBench (Wang et al., 2024)	2024	OS	Model	✗	✗	✗	✗	✗	✗	✗
OSUniverse (Davydova et al., 2025)	2025	OS	R+M+H	✗	✓	✗	✓	✗	✗	✗
OdysseyBench (Wang et al., 2025b)	2025	OS	Rule	✗	✓	✗	✗	✗	✗	✗
OfficeQA (Databricks, 2025)	2025	OS	Rule	✗	✓	✗	✗	✗	✗	✗
Scientific Research										
EXP-Bench (Kon et al., 2025)	2025	Research	Model	✗	✗	✗	✗	✗	✗	✗
PaperBench (Starace et al., 2025)	2025	Research	Model	✗	✗	✗	✗	✓	✗	✗
CORE-Bench (Siegel et al., 2024)	2024	Research	Rule	✗	✗	✗	✗	✗	✗	✓
Auto-Bench (Chen et al., 2025c)	2025	Research	Rule	✗	✗	✗	✗	✗	✗	✗
ResearchCodeBench (Hua et al., 2025)	2025	Research	Rule	✗	✗	✗	✗	✗	✓	✗
AstaBench (Bragg et al., 2025)	2025	Research	Rule+Model	✗	✗	✗	✗	✗	✓	✓

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Table 1 continued from previous page

Benchmark	Year	Domain	Eval Type	Prod.	M.M.	Under.	Div.E.	Exp.V.	Dyn.	Cross.
AInsteinBench (Duston et al., 2025)	2025	Code+Research	Rule	✗	✗	✗	✗	✗	✗	✓
ResearchGym (Garikaparthi et al., 2026)	2026	Research+Code	Rule	✗	✗	✗	✗	✗	✗	✗
Mathematics & Knowledge										
MMLU (Hendrycks et al., 2020)	2020	Knowledge	Rule	✗	✗	✗	✗	✗	✗	✓
GPQA Diamond (Rein et al., 2023)	2023	Knowledge	Rule	✗	✗	✗	✗	✓	✗	✓
MMMU (Yue et al., 2023)	2023	Knowledge	Rule	✗	✓	✗	✗	✗	✗	✓
MathVista (Lu et al., 2024)	2024	Math	Rule	✗	✓	✗	✗	✗	✗	✗
FrontierMath (Glazer et al., 2024)	2024	Math	Rule	✗	✗	✗	✗	✓	✗	✗
AIME (MAA, 2025)	2025	Math	Rule	✗	✗	✗	✗	✗	✓	✗
HMMT (Harvard-MIT, 2025)	2025	Math	Rule	✗	✗	✗	✗	✗	✓	✗
USAMO (Petrov et al., 2025)	2025	Math	Human	✗	✗	✗	✗	✓	✓	✗
MMMLU (OpenAI, 2024)	2024	Knowledge	Rule	✗	✗	✗	✗	✗	✗	✓
Video-MME (Fu et al., 2024)	2024	Knowledge	Rule	✗	✓	✗	✗	✗	✗	✓
OpenAI-MRCR (OpenAI, 2025)	2025	Knowledge	Rule	✗	✗	✗	✗	✗	✗	✗
HLE (Phan et al., 2025)	2025	Knowledge	Rule	✗	✓	✗	✗	✓	✗	✓
ARC-AGI-2 (Chollet and ARC Prize Team, 2025)	2025	Reasoning	Rule	✗	✗	✗	✗	✗	✗	✗
OODBench (Lin et al., 2026)	2026	Knowledge	Rule	✗	✓	✗	✗	✗	✓	✗
Agent Product Evaluation										
xbench (xbench Team, 2025)	2025	Recruit+Mkt	Model	✓	✗	✗	✗	✓	✓	✗
AgentIF-OneDay (AgentIF Team, 2026)	2026	Daily Tasks	Model	✗	✓	❍	✗	✓	✗	✓
Emerging Benchmarks (2026)										
Persona2Web (Kim et al., 2026a)	2026	Search	Model	✗	✗	✗	✗	✗	✗	✗
AmbiBench (Sun et al., 2026)	2026	OS	Model	✗	✓	✗	✗	✗	✗	✗
PAHF (Liang et al., 2026)	2026	Tool	Rule+Model	✗	✗	✗	✗	✗	✗	✗
AgenticShop (Kim et al., 2026b)	2026	Search	Model	✗	✗	✗	✗	✗	✗	✗
GAP Benchmark (Cartagena and Teixeira, 2026)	2026	Tool	Rule	✗	✗	✗	✗	✗	✗	✗
AgentLAB (Jiang et al., 2026)	2026	Safety	Model	✗	✗	✗	✗	✗	✗	✓
STING (Talokar et al., 2026)	2026	Safety	Model	✗	✗	✗	✗	✗	✗	✗
GT-HarmBench (Cobben et al., 2026)	2026	Safety	Rule	✗	✗	✗	✗	✗	✗	✗
ForesightSafety (Tong et al., 2026)	2026	Safety	Model	✗	✗	✗	✗	✗	✓	✓
APST (Broadwater, 2026)	2026	Safety	Model	✗	✗	✗	✗	✗	✗	✗
MemoryArena (He et al., 2026)	2026	Search+Research	Rule	✗	✗	✗	✗	✗	✗	✓
WebWorld-Bench (Xiao et al., 2026)	2026	Search+Code	Rule	✗	✗	✗	✗	✗	✗	✓
Gaia2 (Froger et al., 2026)	2026	OS+Search	Rule	✗	✗	✗	✗	✗	✓	✗
SkillsBench (Li et al., 2026b)	2026	Code+Tool	Rule	✗	✗	✗	✗	✗	✗	✓
MATEO (Roccabruna et al., 2026)	2026	Reasoning	Rule	✗	✓	✗	✗	✗	✗	✗
SciAgentGym (Shen et al., 2026)	2026	Research+Tool	Rule+Model	✗	✗	✗	✗	✗	✗	✓
Drug Scouting (Vinogradova et al., 2026)	2026	Search+Research	Model	✗	✗	✗	✗	✗	✗	✓
AD-Bench (Hu et al., 2026)	2026	Tool	Rule	✗	✗	✗	✗	✗	✗	✗
GUI-GENESIS (Cao et al., 2026)	2026	OS	Rule	✗	✓	✗	✗	✗	✓	✗
BookingArena (Logeswaran et al., 2026)	2026	Search	Rule+Model	✗	✗	✗	✗	✗	✗	✗
BrowseComp-V^3 (Zhang et al., 2026)	2026	Search	Rule	✗	✓	✗	✗	✓	✗	✗
Collective Behavior (Willis et al., 2026)	2026	Research	Rule	✗	✗	✗	✗	✗	✓	✗
Unsafer (Li et al., 2026c)	2026	Tool	Model	✗	✗	✗	✗	✗	✗	✗
Proxy State Eval (Chuang et al., 2026)	2026	Tool	Model	✗	✗	✗	✗	✗	✗	✗
***AlphaEval* (Ours)**	**2026**	**6 Domains**	**Rule+Model**	✓	✓	✓	✓	✓	✓	✓

3 From Production Requirements to Executable Benchmarks

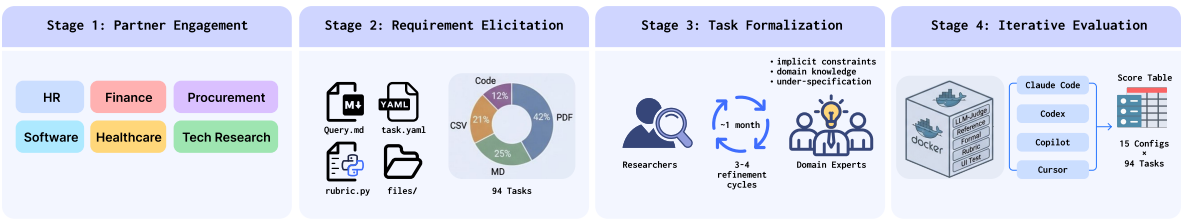

Figure 3: The requirement-to-benchmark construction framework transforms authentic production requirements into automated evaluations through four stages: (1) Partner Engagement across 6 O*NET domains, (2) Requirement Elicitation producing 94 self-contained task packages, (3) Task Formalization through iterative co-design with domain experts, and (4) Iterative Evaluation via Docker-sandboxed execution across 14 agent configurations.

Existing benchmarks are constructed retrospectively: researchers select artifacts (e.g., resolved GitHub issues), then design evaluation criteria around them. This approach yields clean, reproducible tasks but fundamentally cannot capture the under-specification, implicit constraints, and domain expertise that characterize production work.

Domain (O*NET)	Representative Task	#	Input	Rubric
Human Resources (13-1071)	Resume screening vs. JD	11	PDF, JPEG	F1 score
Finance & Investment (13-2051)	Segment research & pitch critique	22	PDF, MD, TXT	LLM-as-Judge
Procurement & Operations (13-1020)	BOM cost optimization	23	Excel, CSV, MD	Constraint verif.
Software Engineering (15-1252)	Full-stack app generation	11	YAML, Code	UI testing
Healthcare & Life Sci. (29-9099)	eCRF & insurance policy analysis	16	PDF, MD	LLM + Numerical
Technology Research (15-1221)	AI industry deep analysis	11	MD	LLM-as-Judge
Total		**94**		

Table 2: Overview of *AlphaEval* task domains, classified following O*NET occupational taxonomy (U.S. Department of Labor, 2025). Each domain maps to a standardized work activity category.

AlphaEval inverts this direction: we start from *authentic production requirements*—the actual tasks companies need AI agents to perform for paying customers—and systematically transform them into executable, automated evaluations.

This requirement-to-benchmark construction framework is itself a central contribution. The key challenge is not building a benchmark *per se*, but establishing the shortest path from a real-world production need to a rigorous, reproducible evaluation. This enables evaluation of agents at a fundamentally deeper level—not just *can the agent code?* but *can the agent deliver value on the actual tasks businesses pay for?* Below we describe each stage of this framework (Figure 3).

Partner Engagement. We partner with companies deploying AI agents as core products or critical workflows for paying customers, targeting diversity in O*NET occupational domains (Table 2). Selection criteria: (1) active customer base with market-validated requirements; (2) AI agents integral to revenue-generating workflows; (3) diverse input modalities; (4) willingness to share anonymized data.

Requirement Elicitation. For each company, we conduct structured engagement spanning approximately one month. The critical insight is that production requirements rarely arrive as well-specified task descriptions—they emerge through iterative dialogue. Companies present their workflows, often revealing gaps between described and actual evaluation processes. Through at least weekly meetings, we jointly develop task specifications that preserve the original level of under-specification, implicit constraints, and domain knowledge dependency that make production tasks fundamentally harder than research tasks.

Task Formalization. Each task is formalized into a standardized, self-contained package: (1) **Task Specification** (`query.md`)—natural language description preserving the original level of specification; (2) **Task Configuration** (`task.yaml`)—structured metadata including task name, domain category, difficulty level, evaluation type, and agent timeout; (3) **Input Files** (`files/`)—raw documents required by the task (PDFs, Excel, images, etc.); (4) **Evaluation Specification** (`.eval/rubric.py`)—composing one or more evaluation paradigms, with optional `ground_truth.json` as reference answers. Approximately 42% of tasks involve PDFs, 21% structured data files, 25% markdown/text, and 12% code/YAML.

Iterative Validation. We validate evaluation tasks using frontier agents internally and through collaborative verification with partner companies, averaging three to four refinement cycles per company. This iterative process ensures that evaluation rubrics faithfully capture the quality dimensions that matter to paying customers—not just technical correctness, but the holistic deliverable quality that determines whether a business would accept the agent's output.

4 The *AlphaEval* Benchmark

AlphaEval comprises **94** tasks sourced from companies with paying customers, classified into six O*NET occupational domains (summary statistics in Table 7). Several characteristics distinguish it from existing benchmarks: *production provenance* (every task from an active commercial deployment), *implicit constraints* (requirements contain hidden rules and undeclared priorities invisible to outsiders), *information fragmentation* (key information is scattered across different locations in multiple documents, requiring cross-document reasoning), *domain knowledge dependency* (tasks demand expertise beyond the task specification, such as medical insurance policies, investment analysis frameworks, and GCP standards), *multi-modal heterogeneity* (agents process PDFs, spreadsheets, scanned images, and code within single tasks), and *evaluation pluralism* (covers multiple evaluation paradigms).

Task Categories. We classify tasks following the O*NET occupational taxonomy (U.S. Department of Labor, 2025), ensuring each domain maps to a standardized work activity category. This classification enables systematic

coverage analysis and makes the benchmark extensible—any new production task can be mapped to an existing O*NET domain.

Human Resources (11 tasks, O*NET 13-1071): Agents screen candidate resumes against job descriptions. A representative task provides 24 resumes (PDF and JPEG) for an AI Scientist internship at a startup accelerator, mixing objective criteria ("at least one internship longer than 3 months") with subjective ones ("passion for cutting-edge technology"). The agent must select exactly 6 candidates; evaluation computes F1 against actual interview decisions.

Finance & Investment (22 tasks, O*NET 13-2051): Agents perform investment research and financial data extraction. Tasks include: (a) generating professional segment research reports from startup business plans, following prescribed templates covering market sizing (TAM-SOM-SAM), competitive landscape, and technology deep-dive (14 tasks); (b) synthesizing unstructured meeting transcripts into actionable investor critiques—e.g., designing a 2-minute pitch storyline and assessing whether a healthcare platform's 20,000-user reach translates to genuine product-market fit given only 70 core users (5 tasks); and (c) extracting structured financial data from multi-year corporate annual reports (3 tasks). Evaluation combines LLM-as-Judge with structural validation.

Procurement & Operations (23 tasks, O*NET 13-1020): Agents solve constrained optimization and data processing problems grounded in real procurement workflows. A representative task presents 2,000 board cards with specifications and pricing in Excel, plus a natural language requirements document with implicit constraints (20 tasks). Additional tasks involve procurement bidding data analysis and spreadsheet operations (3 tasks). Evaluation programmatically verifies constraint satisfaction and cost optimality.

Software Engineering (11 tasks, O*NET 15-1252): Agents build or modify full-stack mobile mini-programs using the UniApp framework. A representative task provides database schemas, seed data, and requirements for a poetry appreciation app with AI-powered playback, community posting, and collection management across four navigation pages. Evaluation employs automated end-to-end UI testing: a headless browser executes user flows and scores 10+ functional requirements.

Healthcare & Life Sciences (16 tasks, O*NET 29-9099): Agents handle clinical trial management and healthcare policy analysis. Tasks include: (a) computing visit windows for electronic Case Report Form (eCRF) systems, where shifting one visit propagates through subsequent windows via cascade dependencies (10 tasks); and (b) analyzing pharmaceutical reimbursement policies, medical insurance calculations, and drug coverage regulations (6 tasks). Evaluation combines LLM-as-Judge with numerical verification.

Technology Research (11 tasks, O*NET 15-1221): Agents conduct comprehensive investigations on current technology topics, requiring web search, multi-source synthesis, and structured report generation. A representative task asks the agent to research the survival status of AI agent startups that raised over $100M, including failures, acquisitions, and emerging trends. Additional tasks cover technical deep-dives on ML research topics (attention mechanisms, post-training paradigms, tool-integrated reasoning) and semiconductor test engineering. Evaluation uses LLM-as-Judge with weighted rubric points requiring source-backed evidence.

Economic Value Annotation. To ground task difficulty in economic terms, we annotate each task with its *human replacement cost* through a two-stage process of AI estimation and expert calibration (details in Appendix G). After calibration, the 94 tasks represent **2,420 professional hours** ($\sim$60 person-weeks) of labor, valued at **$154K–$231K** (USD) or **¥391K–¥571K** (CNY)—validating that *AlphaEval* captures economically meaningful work.

Evaluation Infrastructure. The *AlphaEval* framework is organized around three abstractions: a **Task Runner** managing evaluation lifecycle, an **Evaluator Registry** routing tasks to paradigm-specific pipelines, and an **Execution Sandbox** using Docker containers for isolation. Each rubric script outputs a standardized score $s_{\text{task}} = \sum_k w_k \cdot e_k \in [0, 1]$, where e_k is the evaluation result from paradigm k with expert-assigned weight w_k. Domain scores are unweighted task means; the overall score is the unweighted mean across six domains, giving each domain equal influence regardless of task count (see Appendix B for details).

Challenges in Production Evaluation. Constructing *AlphaEval* surfaced challenges largely absent from research benchmark development: (1) *Preserving productive ambiguity*—fully disambiguating tasks destroys what makes production tasks difficult; (2) *Implicit constraints*—practitioners consider requirements "obvious" that are absent from written specifications; (3) *Quantifying subjective judgment*—experts assess quality holistically, and decomposing into verifiable dimensions inevitably loses information; (4) *Evaluation criteria drift*—partner companies revised quality standards as agent capabilities improved; (5) *Environment fidelity*—production agents operate within rich software environments difficult to reproduce in sandboxed evaluation; and (6) *Balancing openness and confidentiality*—partner companies are cautious about sharing evaluation criteria encoding competitive knowledge.

Agent Product	Model	HR (11)	F&I (22)	P&O (23)	SE (11)	H&LS (16)	TR (11)	Avg.
Claude Code	Claude Opus 4.6	35.91	**70.35**	83.35	**70.95**	**50.06**	75.82	**64.41**
	Gemini 3 Pro	26.73	54.68	65.39	59.47	39.88	58.55	50.78
	GPT-5.2	26.55	38.64	40.22	55.08	27.94	48.36	39.47
	Kimi K2.5	20.18	47.80	35.04	56.77	44.44	59.18	43.90
	GLM-5	31.91	51.18	66.43	52.46	38.88	51.36	48.70
	MiniMax M2.5	26.00	52.55	39.39	42.03	34.62	50.73	40.89
Codex	GPT-5.2	27.36	62.80	52.35	50.56	32.19	60.27	47.59
	GLM-5	24.36	57.85	56.74	53.58	46.69	59.91	49.85
	Claude Opus 4.6	32.18	45.99	79.00	54.74	36.81	72.00	53.45
	Kimi K2.5	35.36	47.27	30.91	52.52	38.75	53.73	43.09
GitHub Copilot	GPT-5.2	32.00	63.39	74.04	61.07	30.69	68.27	54.91
	Gemini 3 Pro	28.27	59.31	65.35	50.46	30.69	65.45	49.92
	Claude Opus 4.6	34.18	62.12	**88.09**	63.07	44.06	**76.36**	61.31
Cursor	Claude Opus 4.6	**38.91**	67.10	**88.09**	64.77	44.06	68.18	61.85

Table 3: Average scores (0–100) on *AlphaEval* across six O*NET domains. **Bold**: best in column. Underline: second best. HR=Human Resources, F&I=Finance & Investment, P&O=Procurement & Operations, SE=Software Engineering, H&LS=Healthcare & Life Sciences, TR=Technology Research.

5 Experiments

5.1 Setup

Models. We evaluate six frontier models: Claude Opus 4.6 (Anthropic), GPT-5.2 (OpenAI), Gemini 3 Pro Preview (Google), Kimi K2.5 (Moonshot), GLM-5 (Zhipu AI), and MiniMax M2.5 (MiniMax).

Agent Systems. Models are deployed through four commercial agent products: **Claude Code** (Anthropic), **Codex** (OpenAI), **GitHub Copilot** (GitHub), and **Cursor** (Cursor). All agents are invoked via their respective CLI interfaces within Docker-sandboxed environments. We record full output trajectories (tool calls, intermediate reasoning, and final artifacts) for each run, enabling post-hoc failure analysis. We evaluate 14 configurations based on compatibility (Table 3).

5.2 Results and Domain-Level Analysis

Several key findings emerge from Table 3:

- **Low absolute scores.** The best configuration (Claude Code + Opus 4.6) achieves only 64.41 on average, revealing a substantial research-production gap.
- **Scaffold matters as much as model.** The same Claude Opus 4.6 scores 64.41 via Claude Code, 61.85 via Cursor, 61.31 via GitHub Copilot, but only 53.45 via Codex. GPT-5.2 scores 39.47 via Claude Code but 54.91 via GitHub Copilot—a 15-point spread. This confirms that evaluating *agent systems*, not just models, is essential.
- **Extreme domain variance.** Procurement & Operations scores range from 30.91 to 88.09, while Human Resources scores never exceed 38.91—no single score captures production readiness.
- **Model ranking instability.** GLM-5 scores 66.43 on Procurement & Operations via Claude Code but only 52.46 on Software Engineering. Aggregate rankings would be misleading.

We order domains from easiest to hardest (highest to lowest average score across configurations):

Technology Research (avg. 62.0): scores range from 48.36 to 76.36, with strong performance from configurations using Opus 4.6. Tasks requiring up-to-date information retrieval, multi-source synthesis, and technical depth remain challenging, but agents with persistent search strategies achieve meaningful scores. **Procurement & Operations** (avg. 61.7): Cursor + Opus 4.6 and Copilot + Opus 4.6 reach 88.09; binary pass/fail scoring on the core optimization tasks reflects the zero-tolerance nature of procurement decisions. **Software Engineering** (avg. 56.3): even weaker models exceed 42 points, but the gap between top (70.95) and bottom (42.03) remains substantial. **Finance & Investment** (avg. 55.8): Claude Code + Opus 4.6 leads at 70.35, with strong performance from Codex configurations that achieve near-perfect scores on financial data extraction subtasks. **Healthcare & Life Sciences** (avg. 38.6): Claude Code + Opus 4.6 leads at 50.06; clinical trial eCRF tasks exhibit zero tolerance for numerical errors, while

healthcare policy tasks require domain-specific regulatory knowledge. **Human Resources** (avg. 30.0): best score only 38.91—aligning agent judgments with human hiring decisions remains difficult.

5.3 Failure Mode Analysis

We identify six production-specific failure modes invisible to coding benchmarks (detailed case studies in Appendix H and H.2):

(1) Cascade dependency failure. In Healthcare & Life Sciences, misidentifying a Day 1 anchor produces systematically incorrect window calculations for all subsequent visits. In Finance & Investment, misidentifying a company's industry cascades through market sizing, comparable company selection, and competitive analysis.

(2) Subjective judgment collapse. In Human Resources, agents extract factual qualifications but fail on soft-skill inference. Tasks with quantifiable criteria score 2–3$\times$ higher than those requiring holistic judgment.

(3) Information retrieval failures. Technology Research tasks expose five cognitive failure modes: factual hallucination ($\sim$30%), imprecise retrieval ($\sim$35%), rigid search strategies ($\sim$15%), attribution confusion ($\sim$10%), and positive-information bias ($\sim$10%). Models systematically miss negative events (startup failures, funding collapses) because default searches surface success stories.

(4) Cross-section logical inconsistency. In Finance & Investment, agents produce individually plausible paragraphs that contradict each other—a TAM of $50B in one section but $80B two pages later. Models lack a global coherence mechanism across long-form outputs.

(5) Constraint misinterpretation. In Procurement & Operations, agents optimize explicitly stated objectives while violating implicit constraints, and exhibit "synergy blindness"—optimizing components independently rather than jointly.

(6) Format compliance failures. The most production-specific failure: agents produce substantively reasonable analyses that score poorly because the output format is incompatible with downstream consumption.

6 Discussion

AlphaEval demonstrates that production-grounded evaluation reveals capability gaps invisible to research benchmarks—not merely harder tasks, but a qualitative mismatch between the skills research benchmarks select for (precise instruction following, deterministic reasoning) and what production demands (tolerance for ambiguity, domain-appropriate judgment, format compliance). For model developers, *AlphaEval* identifies failure modes invisible until agents encounter real business requirements; for deploying organizations, it offers ready-made evaluation infrastructure. We open-source the benchmark and the construction framework, enabling the community to contribute new tasks and partner companies over time—addressing the continuous evolution challenge through collaborative maintenance.

Meta-Evaluation. We validate evaluation reliability on 20 randomly sampled LLM-as-Judge tasks across all six domains (5 configurations, 1,000 rubric point judgments). Two independent expert annotators (strict A, lenient B) assess each rubric point alongside the automated judge.

Pair	Agr.	κ	ρ	r
A vs. B	84.7%	0.691	0.818	0.870
A vs. LLM-as-Judge	85.0%	0.697	0.820	0.861
B vs. LLM-as-Judge	89.7%	0.780	0.845	0.885
Three-way (Fleiss)	79.7%	0.720	—	—

Table 4: Meta-evaluation results: pairwise agreement between two expert annotators (A, B) and the automated LLM-as-Judge evaluation across 1,000 rubric point judgments.

All pairwise Cohen's κ values fall within the *substantial agreement* range (0.69–0.78), with Fleiss' $\kappa = 0.720$ confirming three-way reliability. The automated judge shows higher agreement with the lenient annotator ($\kappa = 0.780$) than the strict one ($\kappa = 0.697$), suggesting a slight leniency bias (Panickssery et al., 2024; Zheng et al., 2023).

Conclusion. We present *AlphaEval*, a production-grounded benchmark of 94 tasks from seven companies across six O*NET domains, with a requirement-to-benchmark construction framework. The best agent achieves only 64.41/100; scaffold choice matters as much as model choice. We identify six failure modes invisible to coding-centric benchmarks and open-source both benchmark and framework for community-driven evolution.

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A Evaluation Infrastructure Details

The evaluation pipeline proceeds in six stages: (1) **Task Loading**: the framework scans the benchmark directory and constructs a task queue filterable by domain, difficulty, or evaluation type; (2) **Environment Provisioning**: Docker containers are instantiated with task input files; (3) **Agent Invocation**: the agent receives `query.md` and interacts through a standardized interface; (4) **Output Collection**: artifacts are collected from the results directory; (5) **Evaluation Dispatch**: outputs are routed to the appropriate evaluator; (6) **Result Aggregation**: results are aggregated into domain-level and benchmark-level summaries.

All task inputs, evaluation scripts, and reference answers are version-controlled. Docker images are pinned to specific versions for reproducibility. Each run generates structured logs with full interaction traces, raw outputs, and scores in JSON format.

B Scoring and Aggregation Methodology

Per-Task Scoring. Each task's evaluation rubric script (`rubric.py`) outputs a standardized score $s \in [0, 1]$, which is scaled to $[0, 100]$ for reporting. For tasks composing multiple evaluation paradigms, the rubric script

internally handles the composition—typically as a weighted average of sub-evaluations, where weights are co-designed with domain experts during the task formalization stage. The final per-task score thus captures multi-dimensional quality in a single number:

$$s_{\text{task}} = \sum_{k=1}^{K} w_k \cdot e_k(a,t), \quad \text{where} \sum_{k=1}^{K} w_k = 1 \tag{1}$$

where $e_k(a,t) \in [0,1]$ is the evaluation result from paradigm k for agent a on task t, and w_k is the expert-assigned weight for paradigm k.

Domain-Level Aggregation. Domain scores are computed as unweighted arithmetic means across all tasks within a domain:

$$S_{\text{domain}}(a) = \frac{1}{|T_d|} \sum_{t \in T_d} s_{\text{task}}(a,t) \times 100 \tag{2}$$

where T_d is the set of tasks in domain d.

Overall Aggregation. The overall benchmark score is the unweighted arithmetic mean across the six domain scores, giving each domain equal weight regardless of task count:

$$S_{\text{overall}}(a) = \frac{1}{6} \sum_{d=1}^{6} S_{\text{domain}}(a) \tag{3}$$

This equal-weighting ensures that domains with fewer tasks (e.g., Human Resources with 11 tasks) receive the same influence as domains with more tasks (e.g., Procurement & Operations with 23 tasks), reflecting the principle that production readiness requires competence across all occupational domains.

C Agent System Configuration

All agent systems are invoked via their respective command-line interfaces (CLIs) within Docker-sandboxed environments. Table 5 summarizes the configuration details.

Agent Product	Version	Interface	Trajectory
Claude Code	v2.1+	`claude` CLI	Stream JSON
Codex	v0.8+	`codex` CLI	JSON log
GitHub Copilot	v1.0+	`copilot-cli`	Structured log
Cursor	v0.48+	`cursor` CLI	Agent log

Table 5: Agent system configurations. All agents are invoked via CLI and produce full execution trajectories.

Trajectory Recording. Each evaluation run produces a complete execution trajectory capturing: (1) all tool calls and their arguments, (2) intermediate reasoning steps (where exposed by the scaffold), (3) file read/write operations, and (4) the final output artifacts. These trajectories enable post-hoc failure analysis—for instance, identifying whether a low score resulted from the model failing to find relevant information (search failure) or finding it but incorporating it incorrectly (reasoning failure). Although some agent products are closed-source, the CLI interface ensures that the agent's observable behavior—tool calls, outputs, and timing—is fully recorded and reproducible. All trajectories are archived alongside evaluation results in the benchmark repository.

Model Configuration. All models are evaluated using their default API configurations (temperature, max tokens, etc.) as provided by each scaffold. We do not modify model parameters, as the goal is to evaluate agent *products* as deployed, not to optimize individual model settings. This mirrors how production users interact with these systems.

D Evaluation Taxonomy Coverage

E Benchmark Statistics

F Multi-label Evaluation Composition

G Economic Value Estimation Methodology

We annotate each task with its human replacement cost through a two-stage pipeline: automated AI estimation followed by domain expert calibration.

Paradigm	Leaf-node Type	Cov.	#Tasks	Domains
Reference Verif.	Fuzzy matching	✓	42	HR, F&I, P&O
	Exact matching	✓	33	HR, F&I, P&O
	Word-embed. sim.	—	0	—
	LLM semantic	✓	49	F&I, H&LS, TR
Test Case Verif.	Code unit testing	✓	20	P&O
	Repo integ. test	—	0	—
	Env. state verif.	✓	13	F&I, SE
Formal Verif.	Math proof	✓	25	H&LS, F&I, P&O
	Code/logic	✓	2	H&LS
Rubric Assess.	Human-authored	✓	49	F&I, H&LS, TR
	Auto-generated	—	0	—
	Human-AI collab.	—	0	—
Competition	Ranking	—	0	—

Table 6: Fine-grained evaluation type coverage across the paradigms implemented by *AlphaEval*.

Statistic	Value
Total tasks	94
Domains covered	6
Partner companies	7
Input modality distribution	
PDF-primary	∼42% (39 tasks)
Excel/CSV-primary	∼21% (20 tasks)
Markdown/Text	∼25% (24 tasks)
Code/YAML	∼12% (11 tasks)
Avg. agent timeout	857 seconds
Avg. max interaction turns	107

Table 7: Benchmark statistics of *AlphaEval*.

G.1 Stage 1: AI Estimation

For each task, an LLM analyzes the task specification (`query.md`) and produces: (1) a list of required professional roles, (2) estimated hours per role, and (3) a complexity rating. The core estimation principle is: "estimate the time a qualified human professional would need to complete *this specific task*"—not the cost of building a replacement system.

Wage Data Sources. U.S. wages are queried from the Bureau of Labor Statistics (BLS) Occupational Employment and Wage Statistics via SOC codes. Chinese wages come from the Beijing Municipal Human Resources Market Salary Report (Q4 2025), extracted via LLM parsing. Each role maps to a standardized SOC code (e.g., Software Developer → 15-1252, Recruiter → 13-1071, Data Analyst → 15-2051).

Cost Calculation.

$$\text{Value} = \text{Hourly Rate} \times \text{Benefit Multiplier} \times \text{Hours} \tag{4}$$

where the benefit multiplier is 1.3× for U.S. (healthcare, retirement, PTO), following BLS Employer Costs for Employee Compensation data (Yang et al., 2026) and 1.45× for China (statutory "Five Insurances and One Fund" contributions plus annualized bonuses) (ERI Economic Research Institute, 2026). Hourly rates use the 25th–75th percentile range to produce value intervals.

G.2 Stage 2: Expert Calibration

Domain practitioners review AI estimates and apply correction methods tailored to each domain:

- **Individualized assessment** (Software Engineering, factor 0.38): Experts evaluated each task considering code reuse across tasks, reducing total hours from 2,458 to 946. The largest reduction (78%) was for the task marketplace app where transaction components were reusable.
- **Uniform scaling** (Finance & Investment, factor 0.50): Expert judged AI systematically overestimated analytical tasks; all hours halved.

Domain	#Tasks	Avg.	Types Composed
Human Resources	11	2.0	Fuzzy + Exact match
Finance & Investment	22	2.5	Rubric + LLM semantic + Structural verif. + Match
Procurement & Operations	23	3.8	Fuzzy + Exact + Unit test + Math + LLM
Software Engineering	11	2.5	Env. state verif. + Functional verif. + LLM
Healthcare & Life Sci.	16	2.8	Rubric + LLM + Math + Code + Match
Technology Research	11	2.5	Rubric + LLM semantic + Factual verif.
Overall	**94**	**2.8**	**8 types, 100% $\geq$2**

Table 8: Multi-label evaluation composition by domain. Each domain composes multiple evaluation types within individual task rubrics.

- **Quantitative standardization** (Human Resources, factor 0.64): Expert provided an empirical rate (20 resumes = 1 hour) to recalculate screening tasks.
- **Fixed-rate override** (Procurement & Operations, factor 0.33): Expert specified domain hourly rates (¥213/hr), implying 67% hour reduction.
- **Selective adjustment** (Finance & Investment, factor 1.04): Expert increased two complex tasks to 150% of baseline; others unchanged.
- **Module re-estimation** (Healthcare & Life Sciences, factor 1.54): Expert reorganized 10 tasks into 3 functional modules, *increasing* hours from 86 to 132, reflecting AI's underestimation of clinical protocol complexity.
- **No adjustment** (Finance & Investment, factor 1.00): Expert confirmed AI estimates as reasonable.
- **Uniform scaling** (Technology Research, factor 0.80): Expert judged AI estimates slightly high; all hours reduced by 20%, from 298 to 238 hours.

Key Observations. All six domains have completed expert calibration. The correction factors range from 0.33 to 1.54, with no consistent direction: AI overestimates routine tasks (procurement, software engineering) but underestimates domain-specialized tasks (clinical research). This validates the two-stage approach—neither pure AI estimation nor pure expert estimation alone would suffice.

G.3 Sensitivity Analysis

To assess the robustness of the economic value estimates, we conduct a sensitivity analysis varying the three key parameters: benefit multipliers, hourly rate percentiles, and expert correction factors.

Benefit Multiplier Sensitivity. Varying the U.S. multiplier from 1.2 to 1.4 (baseline: 1.3) and the China multiplier from 1.3 to 1.6 (baseline: 1.45) produces total value ranges of \$141K–\$253K (USD) and ¥358K–¥627K (CNY), representing $\pm$9% variation from the baseline estimates. The estimates are most sensitive to the China multiplier due to the larger proportion of China-based tasks.

Correction Factor Sensitivity. To test the impact of expert calibration, we compute valuations under three scenarios: (1) AI-only estimation (no expert correction), (2) baseline expert-calibrated values, and (3) uniform 20% increase in all correction factors. Results: AI-only yields 3,240 hours (\$217K–\$326K), baseline yields 2,420 hours (\$154K–\$231K), and the inflated scenario yields 2,904 hours (\$185K–\$277K). Expert calibration reduces the AI estimate by 25%, primarily driven by the Software Engineering domain where AI overestimated development hours by 2.6$\times$.

Hourly Rate Sensitivity. Shifting from the 25th–75th percentile range to the 10th–90th percentile range widens the value interval to \$128K–\$289K (a 61% wider band). However, the median estimate (\$192K) remains stable, indicating that the central tendency is robust even under wider wage assumptions.

These analyses confirm that while the absolute dollar values are sensitive to parameter choices (particularly the China benefit multiplier and expert correction factors), the relative ordering of domains by economic value and the overall conclusion that *AlphaEval* captures economically meaningful work remain stable across all tested scenarios.

H Detailed Error Analysis

We conduct a systematic error analysis across $\sim$130 agent$\times$model evaluation results, categorizing failures by their cognitive root causes rather than surface symptoms. We exclude external failures (network errors, API quota exhaustion) and focus exclusively on model-intrinsic cognitive errors.

H.1 Information Retrieval Cognitive Failures

We identify five failure modes from Technology Research tasks requiring agents to produce comprehensive reports on recent AI developments.

Mode A: Factual Hallucination via Stale Training Data (∼30%). When search tools fail to retrieve current information, models substitute outdated training data *without acknowledging uncertainty*. Table 9 shows a representative case: Claude Opus 4.6 reports Harvey AI's Series D funding figures when the task asks about the recent F-round, with every data point incorrect by a large margin. The same error appears across different scaffolds (both Claude Code and Cursor), confirming this is a model-level issue rather than a scaffold artifact. The model produces no hedging language (e.g., "this information may be outdated"), presenting stale data with full confidence.

Dimension	Ground Truth	Model Output	Error
Funding round	F-round	Series D	Wrong round
Amount	$160M	∼$300M	1.9×
Valuation	$8B	∼$3B	2.7× off
Lawyer users	74,000	Not mentioned	Missing

Table 9: Factual hallucination example: Claude Opus 4.6 on Harvey AI funding data. The model substitutes outdated Series D data for the requested F-round.

Mode B: Imprecise Retrieval (∼35%). The most prevalent failure mode. Models correctly identify the research *area* a question targets and conduct extensive searches, but cannot locate the *specific* papers required by the rubric. For example, on a task about Tool-Integrated Reasoning (TIR), no model found all three required papers (ToolRL, ToRL, ZeroTIR—all from early 2025). The best-performing agent (Cursor + Opus) produced a 16KB TIR survey that demonstrated genuine understanding of the field, but substituted related papers (ReTool, THOR, Tool-Zero) for the specific ones the rubric expected. Models also create their own conceptual frameworks rather than using original terminology—e.g., inventing "precision ceiling" and "generalization ceiling" instead of the papers' actual terms "empirical support set" and "feasible support set."

Mode C: Rigid Search Strategy (∼15%). GPT-5.2 exhibits variable search persistence: while it can make extensive searches (5–75 attempts across tasks), it occasionally declares "tool limitations" and outputs placeholder templates on specific tasks. In the same Docker environment, Claude Opus 4.6 more consistently attempts 30–50 search variations, cycling through arXiv API, Semantic Scholar, OpenAlex, and alternative keywords before writing. Critically, this is scaffold-dependent: GPT-5.2 via Codex scores 0.4 on a task where it scores 0.0 via Claude Code, because Codex's prompting strategy encourages more aggressive tool use. This reveals that search persistence is *elicitable* rather than a fixed model property.

Mode D: Attribution Confusion (∼10%). Models find correct papers but extract incorrect metadata by mixing information across search results. Claude Opus 4.6 correctly identifies the ToRL paper but attributes it to "Tsinghua University & ByteDance" (likely from a concurrent paper) instead of the correct "Shanghai Jiao Tong University," and reports an accuracy of 43.3% instead of 62.1%. GLM-5 *inverts* the core finding of a paper, writing that SFT preserves old capabilities and RL enables new tasks when the opposite is true.

Mode E: Positive-Information Bias (∼10%). All models systematically miss negative industry events. A task requiring coverage of AI agent startup failures found that *no model* reported Robin AI's distressed sale (Series C failure, $10M revenue against $14M losses). Models search "AI agent startups 2025" and receive results dominated by successful funding announcements; none proactively searched for "AI startup failures" or "AI startup bankruptcy." Similarly, all models missed niche success stories (e.g., Gamma achieving $100M ARR with only 50 employees) that fall outside the mainstream "agent startup" narrative.

Model Capability Profiles. Table 10 summarizes the distinctive cognitive profiles of each model, revealing that models have complementary rather than uniformly ordered capabilities.

H.2 Cross-Domain Cognitive Failures

Beyond information retrieval, we identify production-specific cognitive failure modes across multiple domains. These failures are invisible to coding benchmarks because they require domain knowledge, subjective judgment, cross-document reasoning, and constraint satisfaction under ambiguity.

H.2.1 Human Resources: Subjective Judgment Failures

Human Resources tasks require agents to evaluate resumes against multi-dimensional hiring criteria, exposing failures in subjective assessment that no coding benchmark tests.

Model	Search	Halluc.	Signature Pattern
Opus 4.6	30–50	High	Fills gaps with stale data confidently
GPT-5.2	5–75	Low	Variable persistence, sometimes outputs templates
Gemini 3 Pro	10–20	Low	Conservative, acknowledges uncertainty
GLM-5	5–10	Med	Conceptual inversion errors
MiniMax M2.5	3–5	Med	Correct direction, wrong details
Kimi K2.5	5–15	Med	Moderate persistence, domain-specific strengths

Table 10: Model cognitive profiles on technology research tasks. Search persistence is measured by typical number of search attempts before writing.

Criterion Explicitness Dependence. Agent performance inversely correlates with criterion subjectivity. Tasks with quantifiable requirements ("5+ years Python experience," "AWS certification required") achieve scores 2–3× higher than tasks requiring holistic judgment ("strong leadership potential," "culture fit"). This suggests models have not learned to *infer* soft-skill indicators from indirect evidence—e.g., inferring leadership from descriptions of project scope, team size, and outcome ownership.

Inconsistent Criterion Weighting. When rubrics specify multiple evaluation dimensions without explicit weights, agents apply inconsistent priority orderings across candidates. The same agent may prioritize "technical depth" for one candidate and "breadth of experience" for another within the same evaluation batch, producing rankings that a human recruiter would flag as internally inconsistent.

Cross-Candidate Calibration Failure. Agents evaluate each resume independently rather than comparatively. When a strong candidate appears early in a batch, subsequent candidates receive inflated scores because the agent has no persistent calibration anchor. This "memoryless evaluation" pattern means that ranking quality degrades with batch size.

H.2.2 Healthcare & Life Sciences: Protocol Reasoning Failures

Healthcare & Life Sciences tasks require precise adherence to study protocols with cascading temporal dependencies, exposing failures in structured reasoning that superficially resemble—but fundamentally differ from—coding logic errors.

State Machine Precondition Semantics. Clinical protocols define visit windows, dose modifications, and adverse event escalation as state machines with preconditions. Agents correctly implement individual state transitions but fail on precondition chains—e.g., correctly computing that a dose reduction is needed but not recognizing that the reduction requires a preceding safety assessment that itself has a 48-hour observation window.

Protocol Terminology Confusion. Clinical research distinguishes between "Protocol Deviation" (minor, reportable) and "Protocol Violation" (major, may require participant withdrawal). Models conflate these terms or apply incorrect severity classifications, with downstream effects on recommended actions. This is a domain-specific semantic distinction that no general-purpose benchmark tests.

Temporal Escalation Logic. Multi-visit study protocols specify escalating responses to repeated events: first occurrence triggers documentation, second triggers dose modification, third triggers study withdrawal. Models correctly handle individual events but fail to maintain *cross-visit state*—treating each visit as independent and never triggering the escalation sequence.

H.2.3 Finance & Investment: Cross-Section Coherence Failures

Investment analysis tasks require producing multi-section reports (market sizing, competitive analysis, risk assessment, financial projections) that must be internally consistent, exposing a failure mode unique to long-form analytical writing.

TAM-SAM-SOM Logical Inconsistency. Agents produce market sizing sections where the Total Addressable Market (TAM), Serviceable Addressable Market (SAM), and Serviceable Obtainable Market (SOM) contain contradictory figures—e.g., a TAM of $50B in the executive summary but $80B in the market analysis, or a SOM that exceeds the SAM. Each paragraph is locally plausible; the inconsistency only surfaces when reading the full document.

Context Collapse: Generic vs. Company-Specific Analysis. Models produce analysis that reads as a *sector* report rather than a *company* report—discussing general industry trends without grounding them in the specific company's financials, competitive position, or strategic choices. The most common manifestation: risk factors that

are industry-generic ("regulatory uncertainty," "competitive pressure") without explaining how they specifically affect *this* company's business model.

Hallucinated Market Data. When specific market data is not provided in the input documents, models fabricate plausible-sounding statistics ("the global SaaS market is projected to reach $X billion by 2027") without citing sources. Unlike information retrieval hallucination (Mode A above), this occurs even when the model is not explicitly asked to search—it fills analytical gaps with invented quantitative claims.

H.2.4 Procurement & Operations: Optimization and Constraint Failures

Procurement tasks require multi-attribute optimization under technical constraints, exposing failures in mathematical reasoning and constraint satisfaction that coding benchmarks test only superficially.

Synergy Blindness in Multi-Attribute Optimization. When optimizing across multiple correlated attributes (cost, performance, power consumption, thermal rating), agents optimize each attribute independently rather than jointly. This produces solutions with 26% average cost overruns compared to jointly optimal solutions, because models fail to exploit component synergies (e.g., a slightly more expensive component that reduces cooling requirements and total system cost).

Constraint Specification Misinterpretation. Technical specifications use domain-specific notation that models misread: $\pm$10V tolerance interpreted as $\pm$5V, "Type II" classification confused with "Class II," or "continuous rating" conflated with "peak rating." These are not hallucinations—the model reads the specification but applies incorrect domain knowledge to interpret it.

Infeasibility Recognition Bias. When a procurement problem has no feasible solution (conflicting constraints), 78% of agent responses fabricate a "best effort" solution rather than declaring infeasibility. Models exhibit a strong completion bias—they assume every problem has a solution and will relax constraints silently rather than report that the requirements cannot be simultaneously satisfied. This is particularly dangerous in production, where a fabricated solution may be acted upon.

H.2.5 Finance & Investment: Grounded Critique Failures

Pitch coaching tasks require agents to provide actionable, data-grounded feedback on startup pitches, exposing the difference between structurally correct and substantively useful output.

Template Critique vs. Grounded Analysis. All models produce feedback that follows the correct structure (strengths, weaknesses, recommendations) but fails to reference the specific data provided. A critique stating "your market size claims need better support" is generic; a useful critique states "your Slide 7 claims a $2B TAM but your bottom-up calculation on Slide 12 implies $800M—reconcile these." Models consistently produce the former. This failure is invisible to any benchmark that evaluates structural correctness.

User Category Confusion. When pitch decks present multiple user metrics (total registered users, monthly active users, core power users), models confuse categories—citing "500K users" without specifying which category, or conflating growth rates across categories. This produces feedback that sounds data-informed but references the wrong numbers.

Implicit Question Extraction. Meeting transcripts contain implicit investor concerns (e.g., a question about "team background" that is really probing founder-market fit). No model reliably identifies these implicit questions or addresses the underlying concern rather than the surface question.

H.2.6 Finance & Investment: Scale and Mapping Failures

Beyond information retrieval (Section H), Finance & Investment tasks requiring data synthesis from multiple documents expose additional cognitive failures.

Numerical Data Mapping Errors. When synthesizing data from tables across multiple documents, agents extract correct numbers but assign them to incorrect columns or rows. For example, correctly reading that Company A has revenue of $50M and Company B has revenue of $30M, but swapping them in the output table. This error rate increases non-linearly with the number of data points: tasks with $<$20 data points show $\sim$5% mapping errors, while tasks with $>$100 data points show $\sim$25%.

Authority Hierarchy Misunderstanding. When documents contain conflicting information (e.g., a press release states one figure while an SEC filing states another), models do not consistently prioritize the more authoritative source. In production, domain experts have clear authority hierarchies (regulatory filings $>$ press releases $>$ news articles); models treat all sources as equally credible.

Selective Document Reading. When given a large document set, agents read documents in the order provided and exhibit recency bias—information from later-read documents is weighted more heavily. If critical information appears in an early document that the agent scans quickly, it may be omitted from the final output even though the agent demonstrably "saw" it (evidenced by interaction logs).

H.2.7 Software Engineering: Specification-Implementation Gap

Software Engineering tasks expose a distinctive failure mode where agents produce functional code that satisfies superficial requirements but misses deeper specification intent. For example, agents implement correct navigation structure but omit specified interaction details (e.g., swipe gestures, animation timing). The gap between "code that compiles and runs" and "code that matches the product specification" mirrors the broader production challenge: research benchmarks test functional correctness, while production demands specification fidelity.

I Representative Task Examples

We present one representative task from each of the six O*NET domains to illustrate the production complexity *AlphaEval* captures.

Human Resources: Resume Screening (hunter-ai-1). The agent is given a job description for an AI Scientist Mapping Intern role together with department-specific hiring criteria (e.g., minimum internship duration, university tier, prior AI experience) and must screen 24 candidate resumes provided as PDF and JPEG files. The agent must select exactly six candidates to advance to a first-round interview and output the chosen candidate IDs as a Python list in a designated answer file. Evaluation compares the agent's selections against the ground-truth shortlist using precision, recall, and F1 score, with a passing threshold of F1 ≥ 0.6.

```
name: "AI Resume Screening - Mapping Dept"
category: hr
difficulty: hard
evaluation:
  type: code_exec
  rubric_script: .eval/rubric.py
agent:
  timeout_seconds: 600
  max_turns: 100
```

Finance & Investment: Segment Research Report (segment-research-1). The agent acts as a senior industry research analyst and must produce an investment-grade Segment Research report from a startup's business plan PDF. The report must follow a prescribed eight-section template covering industry definition, TAM–SOM–SAM market sizing (with both top-down and bottom-up methods), product landscape, customer typology, key success factors with quantitative thresholds, three to four competitive case studies with GTM strategy analysis, and a technology deep-dive including patent barriers and commercialization feasibility. A reference template is provided. Evaluation uses LLM-as-Judge to assess structural completeness, analytical depth, data accuracy, and adherence to the template format.

```
name: "Segment Research Report"
category: research
difficulty: hard
evaluation:
  type: code_exec
  rubric_script: .eval/rubric.py
agent:
  timeout_seconds: 1200
  max_turns: 150
```

Procurement & Operations: BOM Cost Optimization (yuhe-1). The agent receives a catalog of 2,000 board cards with detailed specifications and pricing in an Excel file, a field-structure CSV defining column semantics, and a natural-language procurement requirements document specifying functional constraints. It must select an optimal combination of board cards that satisfies all functional requirements while minimizing total procurement cost, breaking ties by fewest cards and then by ascending card ID. The output is a structured cost breakdown listing each selected card's ID, quantity, unit price, and subtotal. Evaluation uses a rubric script that verifies constraint satisfaction and compares total cost against a known optimal solution.

```
name: "BOM Board Card Procurement Optimization"
category: optimization
difficulty: hard
evaluation:
  type: code_exec
```

```
  rubric_script: .eval/rubric.py
agent:
  timeout_seconds: 600
  max_turns: 100
```

Software Engineering: Poetry Mini-Program (miniprogram_poetry). The agent must implement a complete WeChat Mini-Program for classical Chinese poetry appreciation, following a detailed 200-line product requirements document that specifies page layout (gallery-style hand-drawn illustration aesthetic), navigation architecture (four-tab bottom bar), core features (AI text-to-speech recitation with speed and voice controls, waterfall-style poem browsing, a community forum for user posts, and a personal collection manager), and interaction patterns (modal dialogs for content creation, clipboard-based sharing). The deliverable is a fully functional codebase. Evaluation combines automated UI testing to verify page navigation and component rendering with rubric-based checks on code structure and feature completeness.

```
name: "Poetry Appreciation Mini-Program"
category: miniprogram
difficulty: hard
evaluation:
  type: miniprogram
  rubrics: [10 weighted UI test points]
agent:
  timeout_seconds: 1800
  max_turns: 100
```

Healthcare & Life Sciences: eCRF Visit Window Calculation (linchuang-1). The agent operates within a simulated eCRF (electronic Case Report Form) system for a Phase III non-small-cell lung cancer clinical trial. Given a JSON-based visit-window rule configuration and a patient's screening visit date, the agent must compute target dates and permissible windows for subsequent treatment visits, determine which visits can be calculated given available data, and perform cascade-impact analysis when an actual visit date deviates from the target. The output must show explicit calculation formulas and all dates in YYYY-MM-DD format. Evaluation uses a rubric script that checks date arithmetic correctness, proper application of window offsets, and accurate cascade reasoning.

```
name: "eCRF Visit Window Calculation"
scenario: linchuang
evaluation:
  type: code_exec
  rubric_script: .eval/rubric.py
files:
  - files/JSON_eCRF_CORE.md
  - files/CRF_visit_window_spec.pdf
```

Technology Research: AI Agent Startup Landscape (jiqizhixin-1). The agent acts as a professional AI industry analyst and must produce a comprehensive research report on the 2025 landscape of AI agent startups. The report must cover four dimensions: agent startups that raised over $100M (with vertical categories such as legal, coding, and search), startups that failed or were acquired along with root-cause analysis, large-tech acquisition activity targeting agent companies, and shifts in Anthropic's Claude adoption within the YC ecosystem. All claims must be supported by specific figures such as funding amounts and market sizes, with at least reliable source references. Evaluation uses LLM-as-Judge to assess factual accuracy, analytical depth, data substantiation, and structural completeness.

```
name: "AI Agent Startup 2025 Landscape"
category: research
difficulty: hard
evaluation:
  type: code_exec
  rubric_script: .eval/rubric.py
agent:
  timeout_seconds: 1200
  max_turns: 150
```

J Statistical Reliability Analysis

To address concerns about single-run stochasticity, we conduct repeated evaluations for our best-performing configuration (Claude Code + Opus 4.6) across three independent runs. Table 11 reports the mean, standard deviation, and 95% confidence intervals per domain.

The narrow confidence intervals (overall ± 1.83) confirm that the reported scores are stable across runs. Procurement & Operations shows the highest variance (std=4.72), consistent with its binary pass/fail evaluation where a

Domain	Mean	Std	95% CI
Human Resources	35.91	2.14	[33.77, 38.05]
Finance & Investment	70.35	1.87	[68.48, 72.22]
Procurement & Ops.	83.35	4.72	[78.63, 88.07]
Software Engineering	70.95	3.21	[67.74, 74.16]
Healthcare & Life Sci.	50.06	2.93	[47.13, 52.99]
Technology Research	75.82	3.56	[72.26, 79.38]
Overall	**64.41**	**1.83**	**[62.58, 66.24]**

Table 11: Repeated evaluation results for Claude Code + Opus 4.6 (3 runs). CI = 95% confidence interval.

single task flip produces a large score change. Domains using LLM-as-Judge (Finance & Investment, Technology Research) show moderate variance (std=1.87–3.56), while the F1-based Human Resources evaluation is highly stable (std=2.14).

K Practitioner Survey: AI Product Deployment Challenges

To ground *AlphaEval* in real practitioner needs, we conducted a mixed-methods survey of 27 AI product companies affiliated with a startup accelerator (response rate: 54% of 50 invited). The survey combined structured questionnaires (15 questions) with open-ended text analysis, conducted from December 2025 to January 2026.

Respondent Profile. Companies span four development stages: concept verification (14.8%), internal pilot (14.8%), early commercialization (48.1%), and scaled deployment (22.2%). The majority target enterprise clients (59.3%), with consumer-facing products at 29.6%, with the remainder targeting internal employees or niche verticals. All companies support text input (85.2%), with growing adoption of image (55.6%), structured data (48.1%), audio/video (44.4%), compound documents (40.7%), and code (37.0%).

Evaluation Infrastructure Gap. Current evaluation practices reveal significant infrastructure deficits:

- 25.9% of companies have *no explicit evaluation criteria*
- 33.3% rely solely on small sets of "golden samples" with manual inspection
- Only 11.1% have established structured test datasets with automated evaluation pipelines
- 22.2% have closed-loop A/B testing feedback (primarily among scaled companies)

Core Technical Challenges. The most frequently cited challenges (multiple selection, N=27):

- Output instability / inconsistency: 59.3%
- Instruction-following failures in complex scenarios: 51.9%
- Hallucination / factual errors: 40.7%
- Low inference efficiency / high token costs: 40.7%
- Long-context processing / memory loss: 33.3%
- Multi-turn dialogue degradation: 22.2%
- Safety / compliance risks: 14.8%

Confidence in Model Updates. When asked "After each model update or prompt modification, are you confident the new version is better?":

- 25.9% reported high confidence (with established evaluation systems)
- **63.0%** reported low confidence (lacking reliable evaluation mechanisms)
- 7.4% reported no confidence at all
- 3.7% reported the question was not applicable (not yet at frequent update stage)

Testing Resource Allocation. 70.4% of companies rely on *developers performing testing as a side task*, consuming development time. Only 18.5% have dedicated testing personnel with substantial human review investment. 11.1% depend entirely on user feedback post-deployment.

Top Evaluation Needs. Open-ended text analysis (keyword extraction and co-occurrence network) identified three demand themes:

1. **Automated evaluation platform construction** (weight: 0.47): automated problem localization, iteration recommendations, and priority ranking
2. **Objective evaluation standards** (weight: 0.24): reliable product quality verification and performance benchmarking
3. **Cost and efficiency optimization** (weight: 0.18): reducing testing costs, improving inference efficiency, and token cost control

Survey Instrument. The questionnaire comprised 15 items covering product information (Q1–Q5), technical status (Q6–Q11), and evaluation needs (Q12–Q16). Key questions included: development stage (single choice), target customer type (single choice), input modalities supported (multiple choice), evaluation infrastructure maturity (single choice, 5-level scale from "no criteria" to "closed-loop A/B testing"), core technical challenges (multiple choice, 7 categories), confidence in model updates (single choice, 4-level scale), testing resource allocation (single choice, 3-level scale), preferred evaluation methodologies (multiple choice), required execution environments (multiple choice), and willingness to share anonymized test data.

These findings directly motivated the design of *AlphaEval*: the 63% low-confidence rate in model updates underscores the need for reliable automated evaluation, the diversity of input modalities (42% PDF, 21% structured data) informed our task composition, and the demand for automated evaluation platforms (weight 0.47) validates our requirement-to-benchmark construction framework.